Wireless Communication

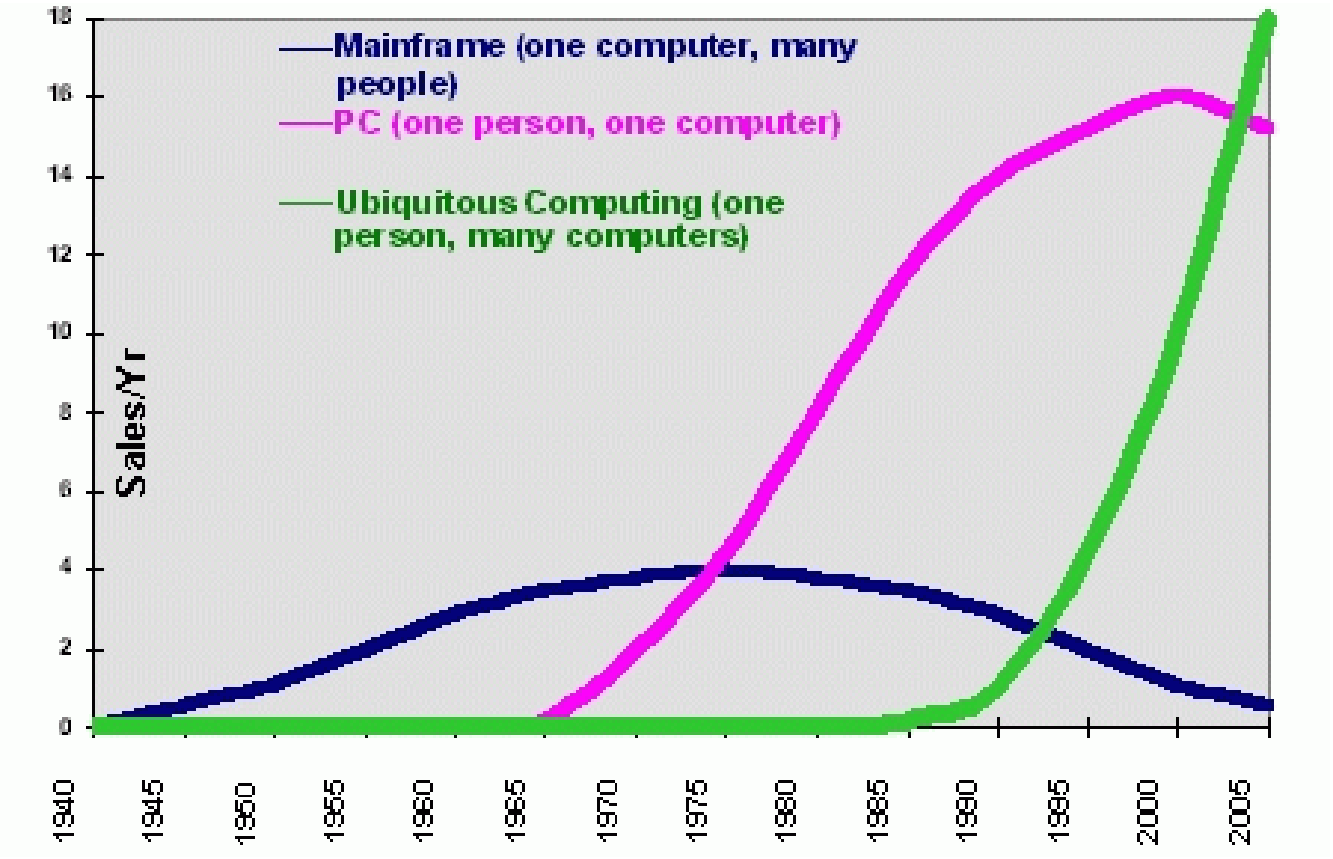
- Basics on Wireless Communications
- Multihop Wireless Networks
- Car-to-Car Communication

Course Mechanics

- Course materials
  - Ad-hoc Wireless Networks: Architectures and Protocols by Murthy and Manoj (Prentice-Hall)
  - Ad-hoc networks slide by Nitin Vidya

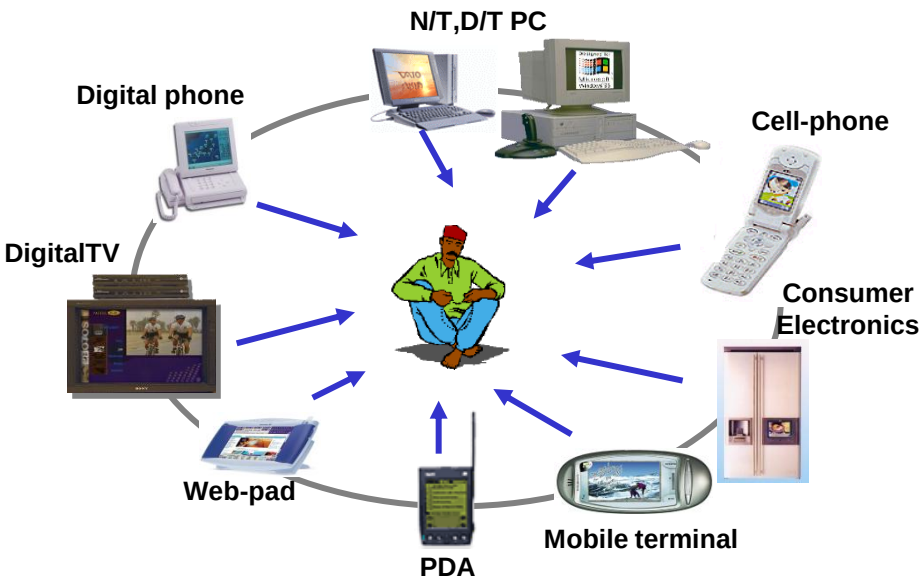
Ubiquitous Computing

- Major trends in computing: Alan Daniel (GIT)



Ubiquitous Computing

- Wireless networked processors embedded in everyday object
- Autonomous and transparent interaction among devices
- Context-aware computing



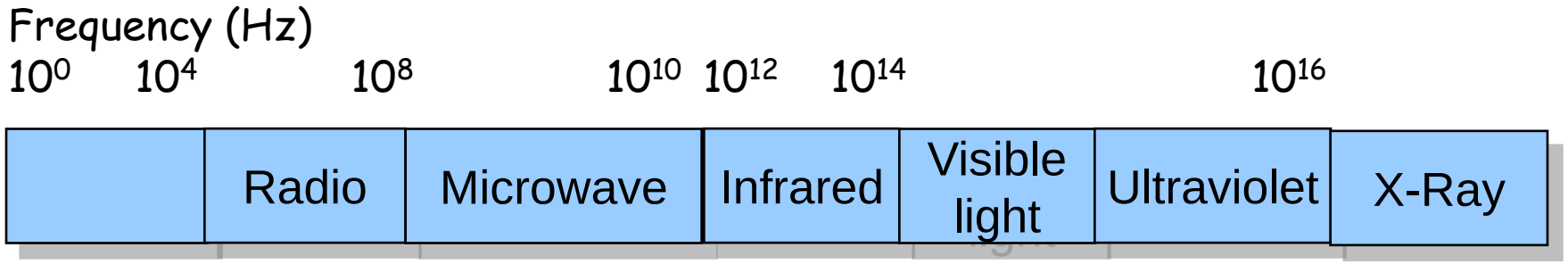
Wireless Communication

- Any form of communication through the air interface

- Characteristics of electromagnetic wave
  - Frequency (f): the number of cycles per second of the wave
  - Wavelength (λ): the distance between two consecutive maxima (or minima) in the wave
  - Period (p): $p = 1/f$
  - Propagation speed (c): $c = \lambda * f$

Wireless Communication

- Electromagnetic wave frequency spectrum



Frequency Bands

Band name	Frequency	Wave length	Applications
ELF	30 – 300 Hz	10000 – 1000 Km	Powerline frequency
VF	300 – 3000 Hz	1000 – 100 Km	Telephone communication
VLF	3 – 30 KHz	100 – 10 Km	Marine communication
LF	30 – 300 KHz	10 – 1 Km	Marine communication
MF	300 – 3000 KHz	1000 – 100 m	AM broadcasting
HF	3 – 30 MHz	100 – 10 m	Long-distance ship/aircraft comm
VHF	30 – 300 MHz	10 – 1 m	FM broadcasting
UHF	300 – 3000 MHz	100 – 10 cm	Cellular telephone
SHF	3 – 30 GHz	10 – 1 cm	Satellite comm, microwave links
EHF	30 – 300 GHz	10 – 1 mm	Wireless local loop
Infrared	300GHz – 400THz	1mm – 770nm	Consumer electronics
Visible light	400 – 900 THz	770nm – 330nm	Optical communication

Radio Propagation Mechanisms

- Direct propagation

- Reflection
  - When the propagating radio wave hits an object which is large compared to its wavelength, the wave gets reflected by the object
  - Reflection causes a phase shift of 180 degrees of the original wave

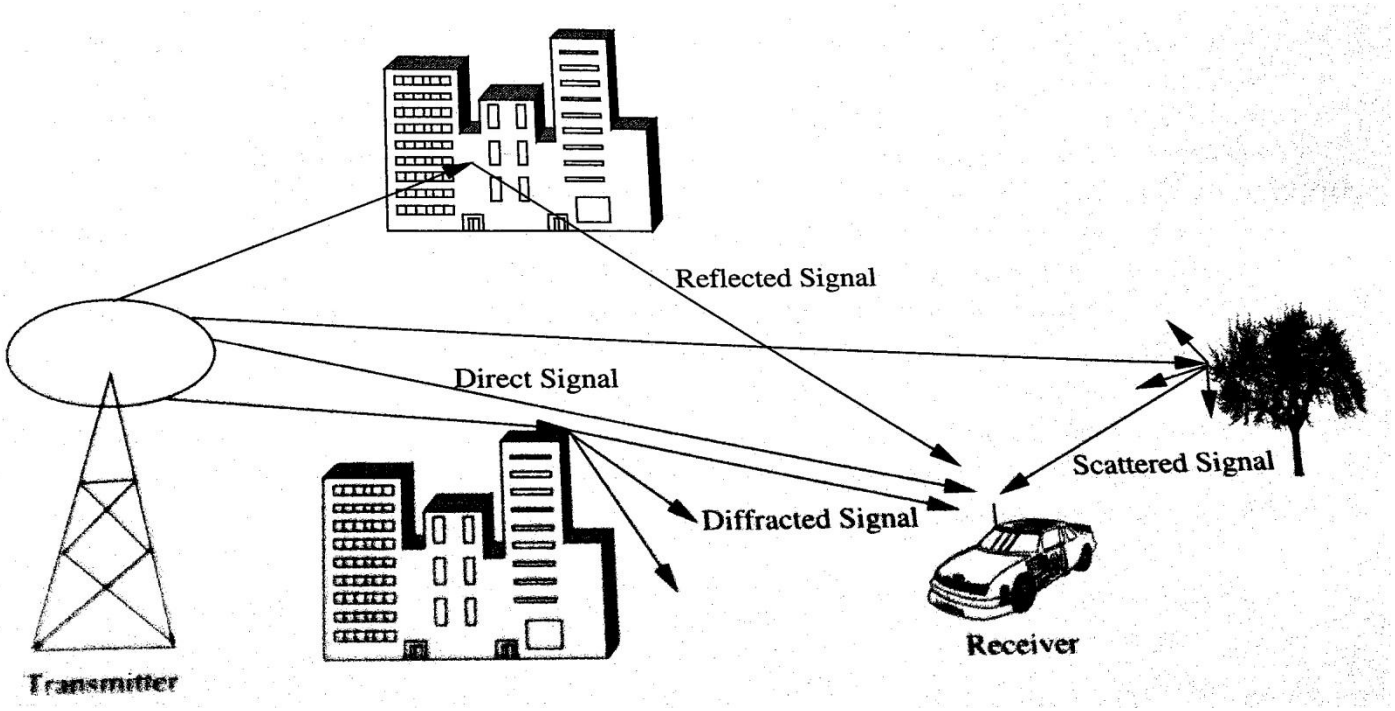
Radio Propagation Mechanisms

- Diffraction
  - When the propagating radio wave hits an object, the wave bends at the edge of the object
  - The lower frequency waves diffracting more

- Scattering
  - When the propagating radio wave encounters an object with dimensions small compared to its wavelength, the wave scatters

Radio Propagation Mechanisms

- Received signal by the receiver contains
  - Direct-path component plus : directly propagated signal
  - Multi-path components : reflected, diffracted, scattered signals



Characteristics of the Wireless Channel

- Path loss
  - The signal power reduces as the distance increases : the signal strength inversely proportional to the square of the propagation distance

- Fading
  - Fluctuations in signal strength
  - Fast fading : the multiple signal paths may sometimes add constructively or sometimes destructively at the receiver
  - Slow fading : occurred by the signal absorbed partially by an object encountered during the propagation; the duration of the fade may last for several seconds or minutes

Characteristics of the Wireless Channel

- Interference
  - Adjacent channel interference : avoid by introducing guard bands
  - Co-channel interference (narrowband interference) : avoid by directional antennas, or dynamic channel allocation mechanisms

Characteristics of the Wireless Channel

- Transmission rate
  - On a noiseless channel : Nyquist's theorem
  - On a noisy channel : Shannon's theorem

$$C = 2 \times B \times \log_2 L \ \text{(bps)}$$

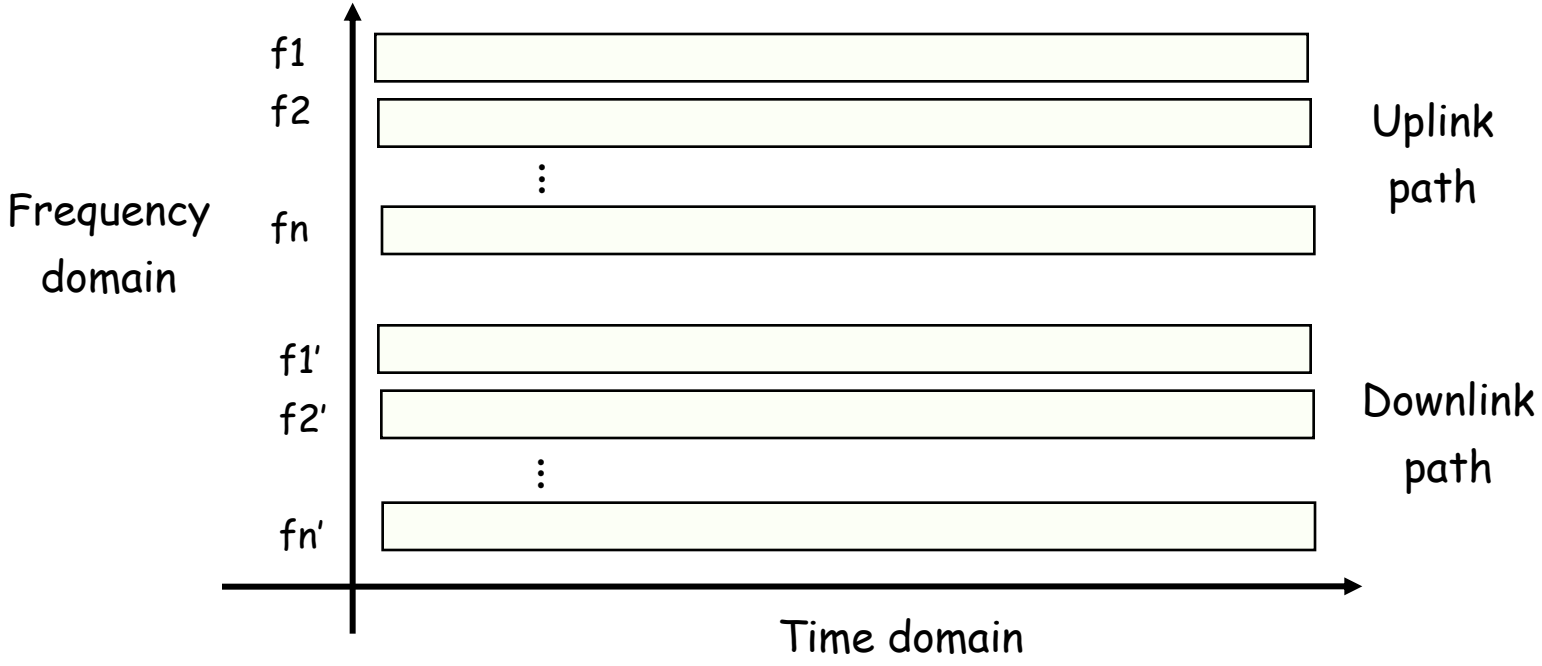
B : bandwidth
L : # of discrete signal levels used

$$C = B \times \log_2 (1 + SNR)$$

SNR (signal to noise ratio)
$= 10 \log_{10} (S/N)$

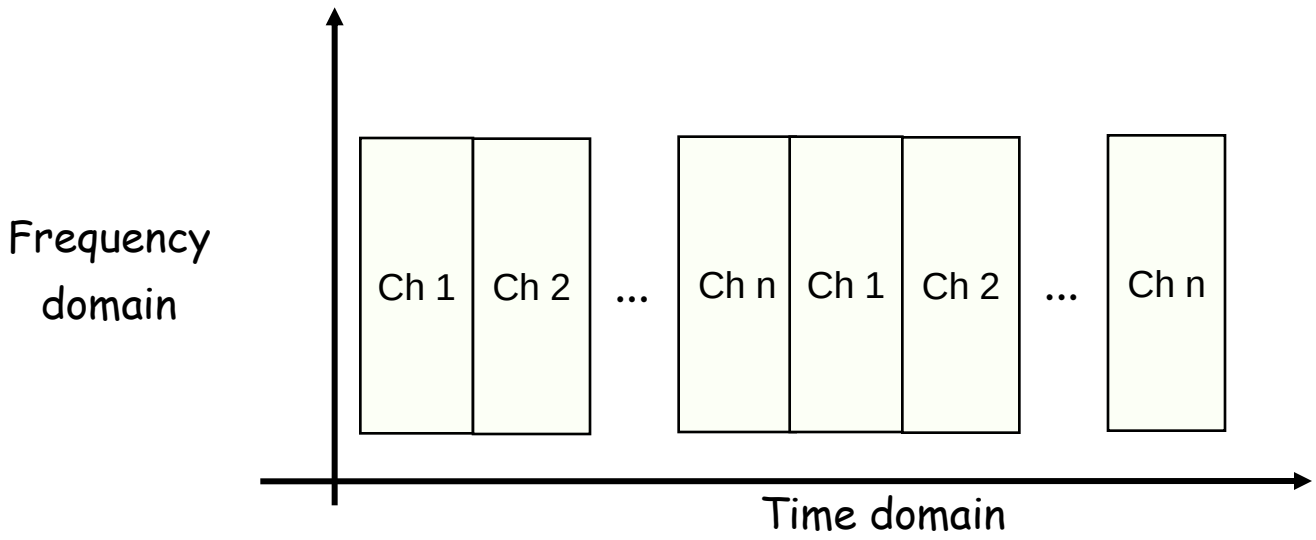
Multiple Access Techniques

- FDMA (Frequency Division Multiple Access)
  - Divides the available radio frequency bandwidth into multiple frequency bands (channels)
  - Needs guard bands between adjacent channels



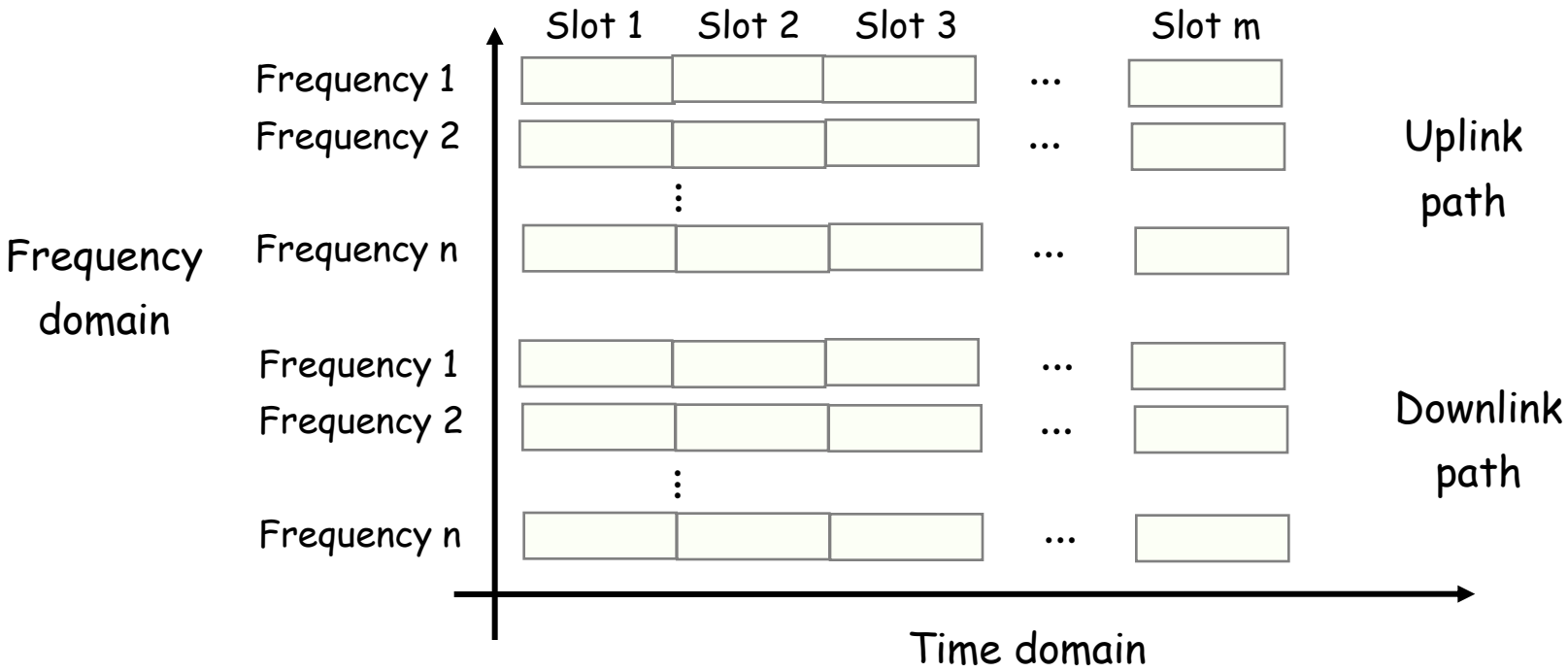
Multiple Access Techniques

- TDMA (Time Division Multiple Access)
  - Divides a frequency band into multiple time slots (channels)
  - TDMA frame : a set of periodically repeating time slots
  - Perfect synchronization is required between the sender and the receiver



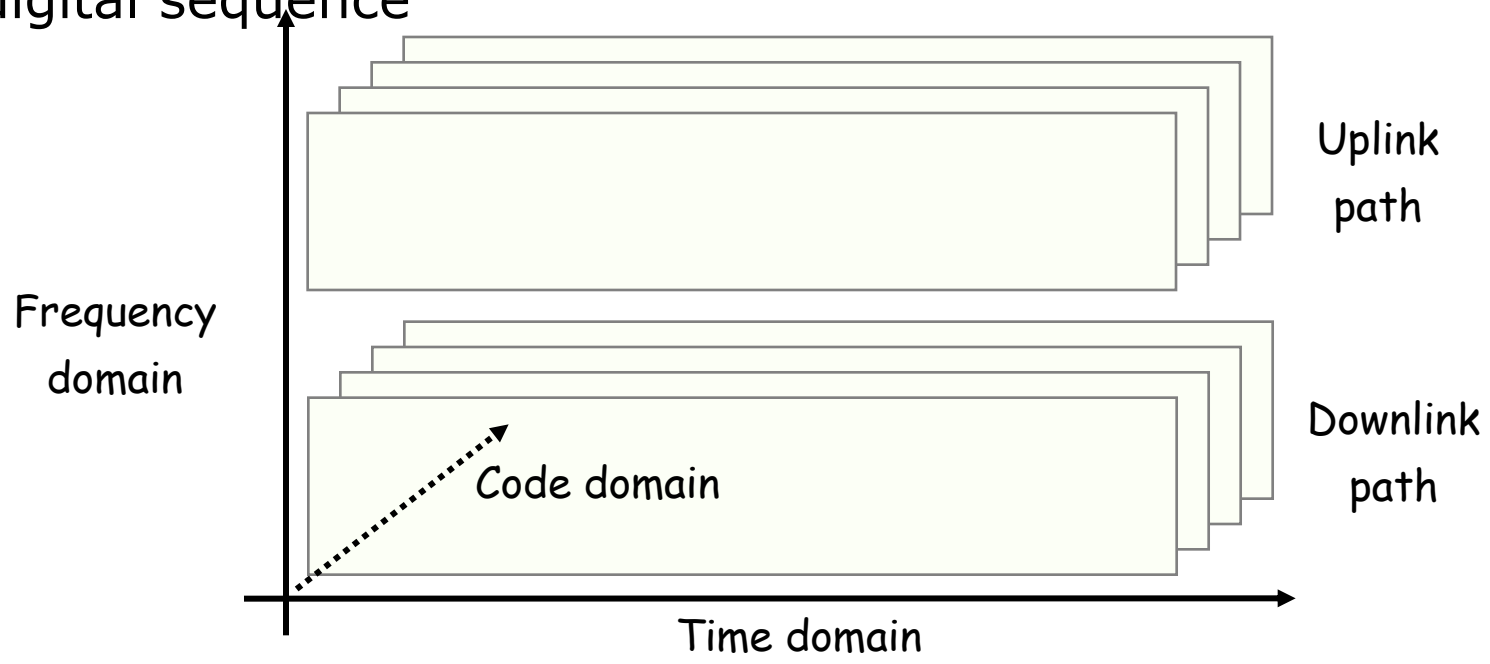
Multiple Access Techniques

- TDMA (Time Division Multiple Access) – cont.
  - Duplex communications : FDD-TDMA (uplink/downlink channels are on two different frequency bands), TDD-TDMA (uplink/downlink channels are on the same frequency band)



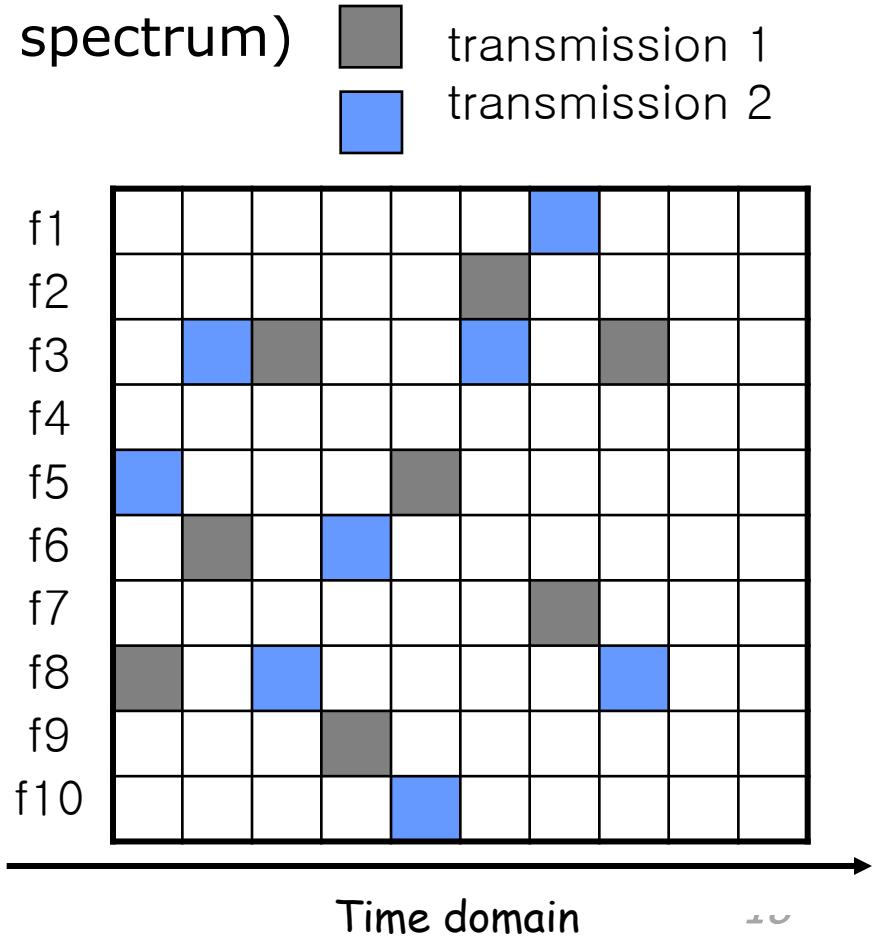
Multiple Access Techniques

- CDMA (Code Division Multiple Access)
  - Uses spread spectrum technique : a narrowband signal (data) is spread over the wideband spectrum
  - Every channel shares the entire spectrum
  - Individual conversations are encoded with a pseudo-random digital sequence



Multiple Access Techniques

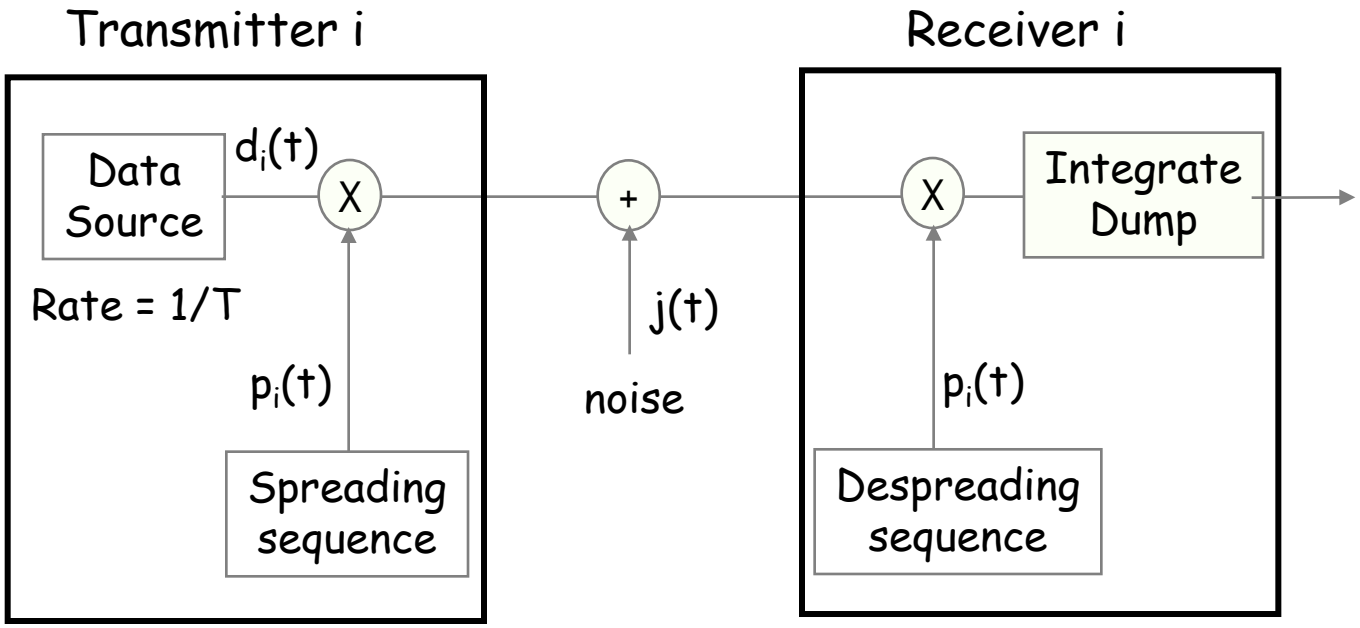
- CDMA (Code Division Multiple Access)
  - FHSS (frequency hopping spread spectrum)
  - DSSS (direct sequence spread spectrum)



- FHSS CDMA
  - Each transmission switches across multiple narrow-band frequencies in a pseudo-random manner: frequency hopping

Multiple Access Techniques

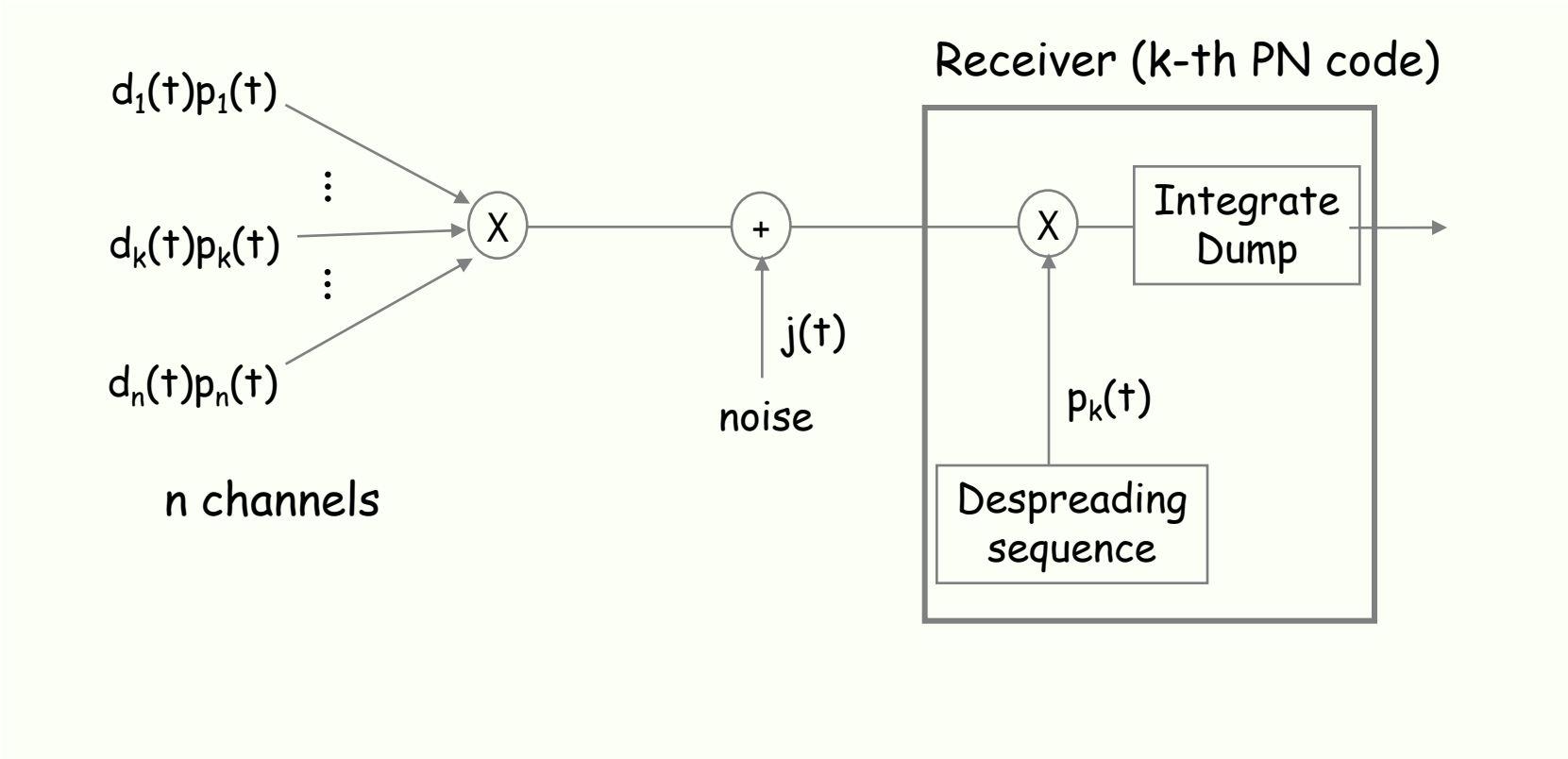
- DSSS CDMA



$p_i(t)$: spreading code (chip code, PN code) for transmission i
m times higher frequency signal than the data signal

Multiple Access Techniques

- DSSS CDMA



Multiple Access Techniques

- DSSS CDMA
  - Received signal at the receiver : r(t)

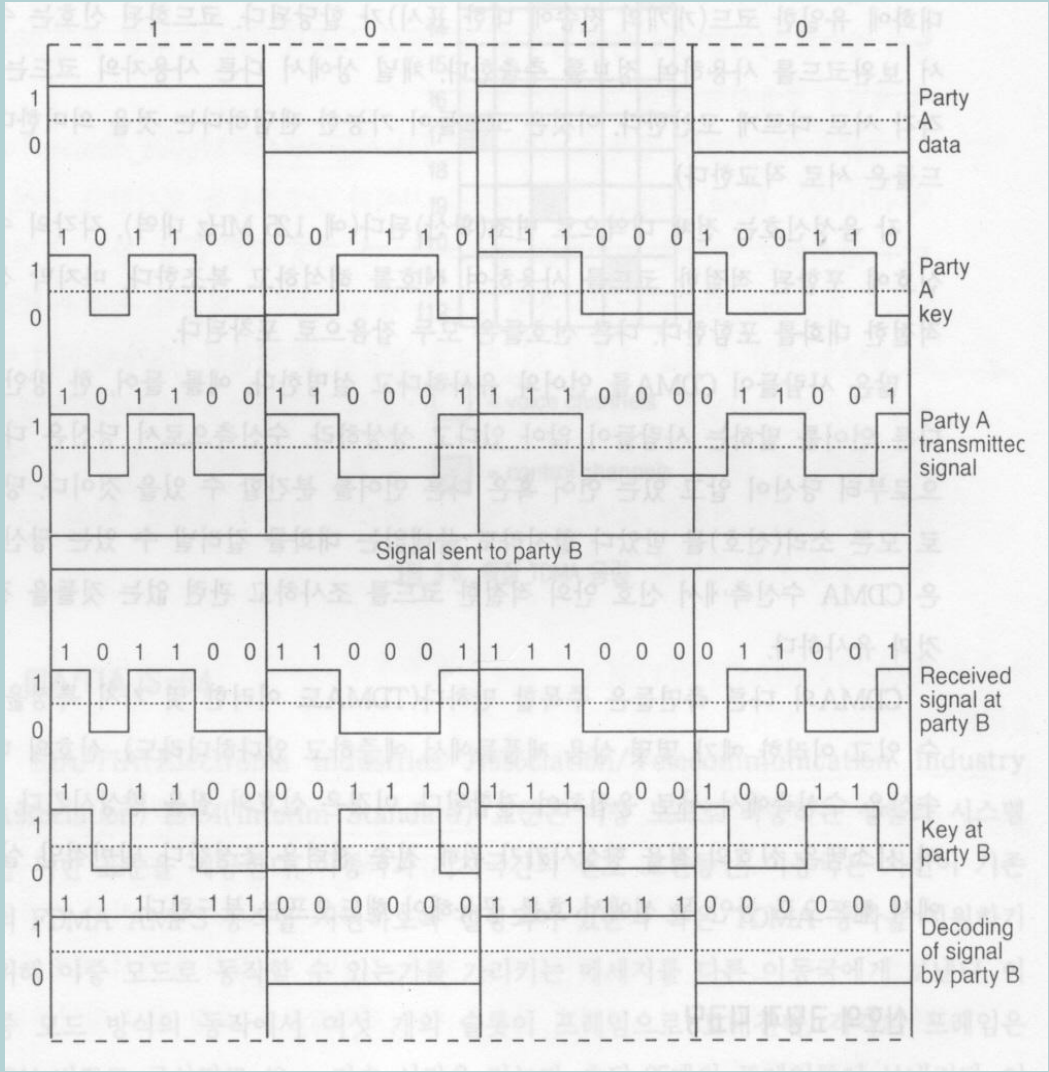
$$r(t) = d_1(t)p_1(t) + d_2(t)p_2(t) + \ldots + d_n(t)p_n(t) + j(t)$$

  - De-spreading at receiver k

$$u_k(t) = r(t)p_k(t)$$
$$= d_1(t)p_1(t)p_k(t) + \ldots + d_k(t)p_k(t)p_k(t) + d_n(t)p_n(t)p_k(t) + j(t)p_k(t)$$
$$u_k(t) = d_k(t)p_k(t)p_k(t) + j(t)p_k(t)$$

$p_i(t)p_j(t) = 0$ if $i \neq j$ ($p_i(t)$ and $p_j(t)$ are orthogonal)

DSSS CDMA


1 0 1 0
Party A data
1 0 1 1 0 0 0 0 1 1 1 0 1 1 1 0 0 0 1 0 0 1 1 0
Party A key
1 0 1 1 0 0 1 1 0 0 0 1 1 1 1 0 0 0 0 1 1 0 0 1
Party A transmittec signal
Signal sent to party B
1 0 1 1 0 0 1 1 0 0 0 1 1 1 1 0 0 0 0 1 1 0 0 1
Received signal at party B
1 0 1 1 0 0 0 0 1 1 1 0 1 1 1 0 0 0 1 0 0 1 1 0
Key at party B
1 1 1 1 1 1 0 0 0 0 0 0 1 1 1 1 1 1 0 0 0 0 0 0
Decoding of signal by party B

Multiple Access Techniques

- SDMA (Space Division Multiple Access)
  - Uses directional transmitters/antennas to cover angular regions
  - The different areas can be served using the same frequency band
  - Enhances frequency spectrum reuse

Wireless vs. Wired

- Noisy and time-varying channel
  - High BER: wired (about 10^{-9}), wireless (about 10^{-4})
  - Environmental conditions affect transmission
  - Mobility effect
- Shared medium
  - Signal interference
  - Needs effective ways of sharing wireless medium
- Bandwidth limited
  - Regulations on the usage of RF bands
  - ISM bands for unlicensed use: 902 – 928 MHz, 2.4 – 2.5 GHz, 5.725 – 5.875 GHz

Wireless vs. Wired

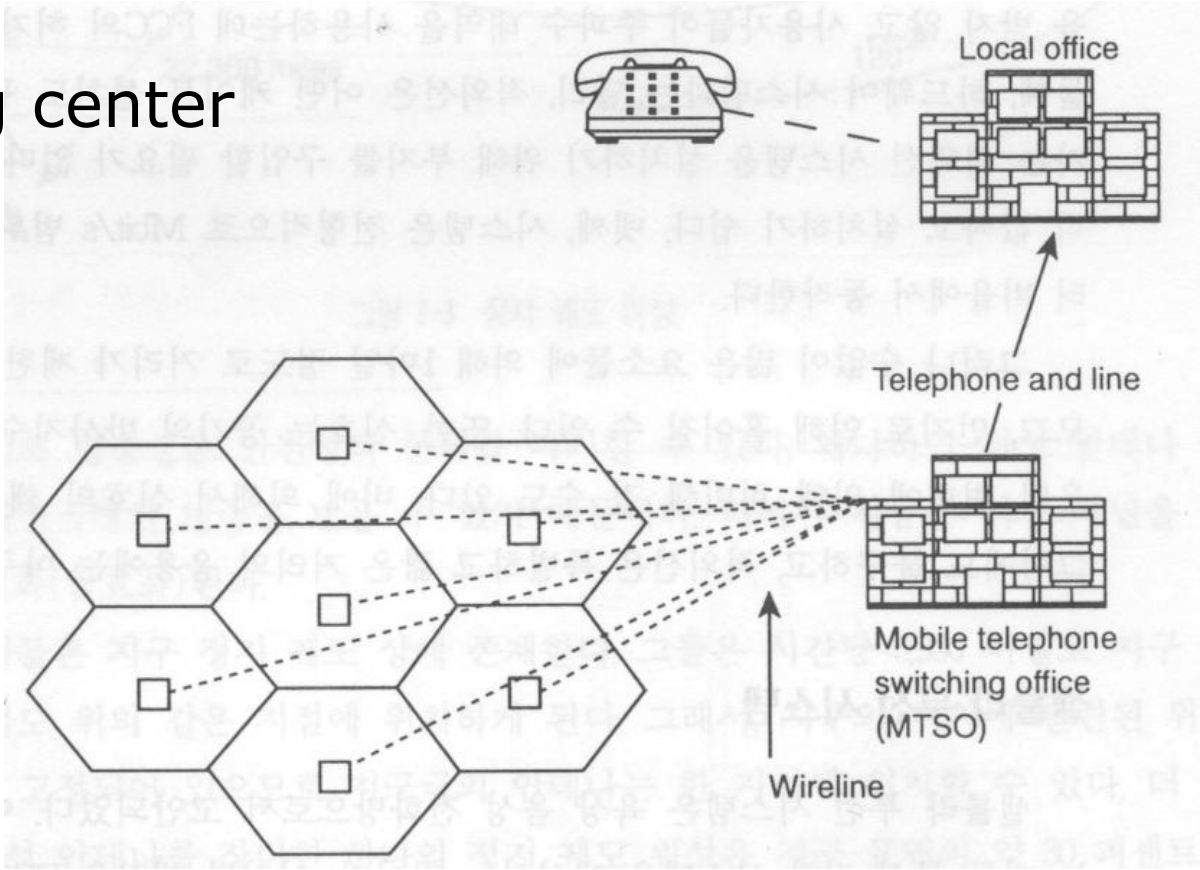
- Communication via broadcasting
  - Transmitted signals can be received by an arbitrary number of users within a coverage area
- Bandwidth scarcity
  - It needs enough power to reach destination node
  - It needs to limit signal power to minimize signal interference and battery usage
  - Bandwidth is a scarce resource: needs to provide means for fair and efficient utilization of available bandwidth among transmitting nodes

Network Topologies

- Types of network topologies
  - Centralized
  - Distributed
  - Hybrid
- Centralized topology
  - Communication among nodes through hub (base station)
  - Base station controls nodes and monitors transmissions from each node
  - Base station manages access by nodes to network's bandwidth
  - (e.g) cellular telephony, WLANs, …

Centralized Topology : Cellular System

- Base station
- Mobile switching center



Centralized Topology

Advantages

- Efficient use of transmit power due to relaying by BS
- Power control is possible for nodes to minimize interference and conserve battery
- Efficient use of bandwidth due to central management by BS
- BS can provide connection to backbone network
- Nodes can be made simple: complexity is in BS
- BS can provide common timing reference

Centralized Topology

Disadvantages

- Single point of failure
- Delay due to the relay by BS
- Cannot cover wide areas where connections exceed range of single link
- Requires significant infrastructure setup
- Not suitable for ad-hoc networks

Distributed Topologies

- De-centralized (peer-to-peer) topology

- Multi-hop wireless network
  - Some nodes cannot reach directly: needs intermediate relay nodes to reach destination
  - Cannot guarantee connectivity of all nodes
  - (e.g) ad-hoc networks, wireless mesh networks, wireless sensor networks

Distributed Topology : Fully-connected

- No single point of failure
- No store-and-forward delay
- No routing: complexity of nodes reduced

- Disadvantages
  - Performance degradation in large networks
  - Near-far problem: transmitters operating at high power levels (to reach far station) will interfere with unintended receivers in close proximity

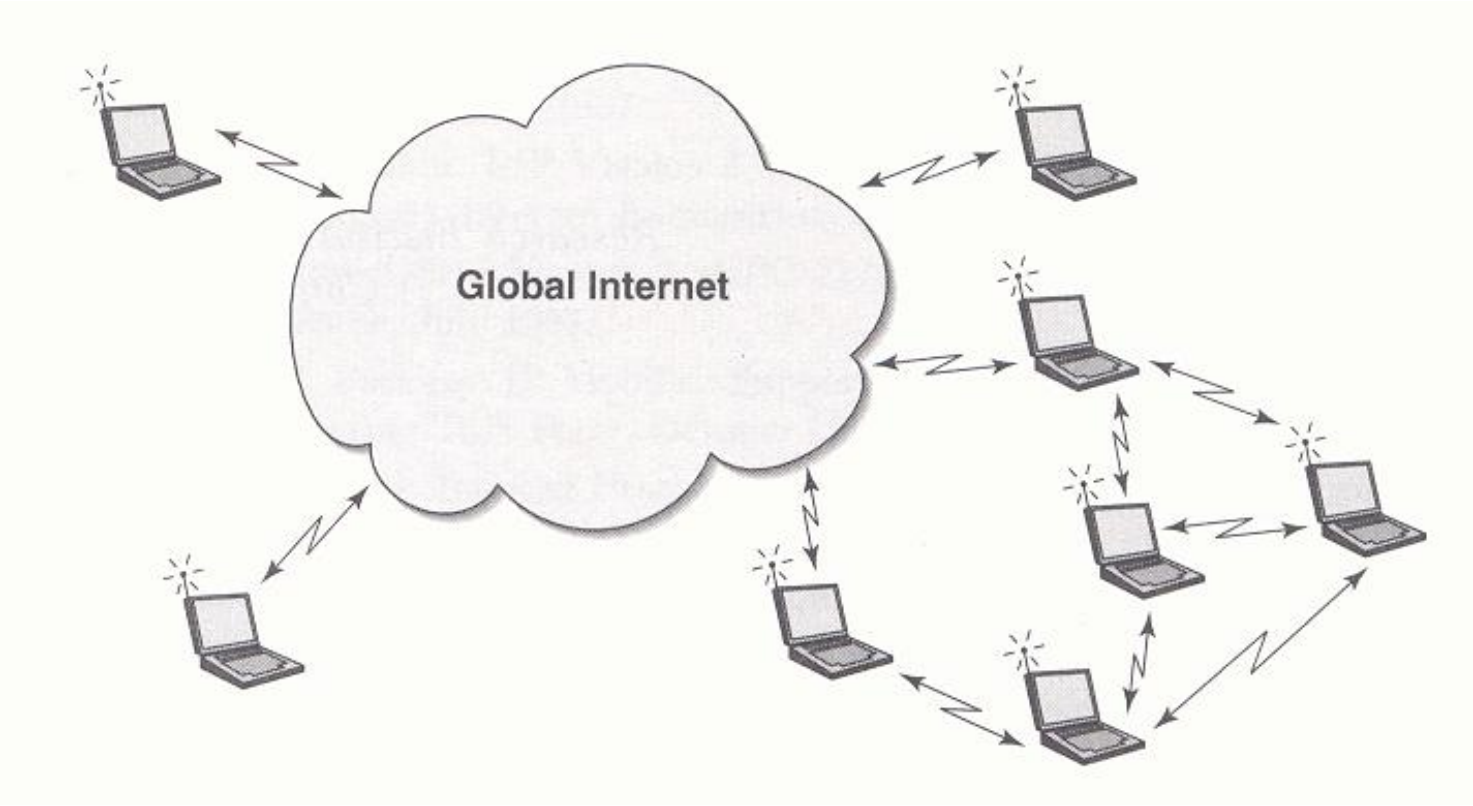
Distributed Topology : Multi-hop

- More efficient energy use
- Communication without infrastructure is possible:
  - Disaster areas, military communication, etc.
  - (e.g.) ad-hoc networks, wireless sensor networks

- Disadvantages
  - Needs complex routing to reach destination : node mobility makes routing more complex
  - More delay due to multiple hops of store-and-forwards
  - No central timing or power control authority

Distributed Topology : Ad-hoc Network

- Ad-hoc network



Types of Wireless Networks

- Wireless LANs
  - WiFi / IEEE 802.11
  - Two modes
  - Centralized: wireless local area data networks
  - Peer-to-peer: MAC/PHY for ad-hoc networks

- Wirelss PANs
  - IEEE 802.15.1: Bluetooth
  - short distance
  - low power

Types of Multihop Wireless Networks

- Ad-hoc networks
  - Multi-hop peer-to-peer networks
  - Hybrid networks
  - Unicast, multicast, broadcast communications

- Wireless sensor networks
  - Different applications use different network topologies
  - Converge-cast, unicast, multicast, broadcast communications

Mobile Ad-hoc Networks (MANETs)

- Definition
  - A collection of wireless mobile nodes forming a temporary network without any centralized administration
- Nodes are mobile → have dynamic network topology
- Have no centralized control or fixed infrastructure : infrastructure-less network
- Examples
  - Conference meetings
  - Emergency relief situations
  - Military communications
  - Wearable computers
  - Sensor networks

MANET

- Limiting communication range of mobile nodes
  - Reduces battery drain
  - Reduces channel interference → enables spatial reuse of limited bandwidth → increased network capacity

- Each node is a packet source or sink, and also is a relay node (router)

MANET

- Dynamic links due to mobility of nodes
  - Broken links must be updated when a node moves out of communication range with another node
  - New links must be formed when a node moves into communication range with another node
  - Routes must be updated based on this information

- Routing is complex → routing in MANET is a big issue

Issues in MANET

- Routing performance
  - Routes change over time due to node mobility
  - Finding optimal path (short delay path, minimum energy path, …) is difficult
  - Avoiding lots of route maintenance overhead

- MAC protocols
  - Provides broadcast communication channel
  - Time-varying channel
  - Power saving operation : nodes should sleep to reduce energy drain

Issues in MANET

- Providing quality of service is difficult
  - High BER
  - Node mobility → broken link → route update
  - Hidden node problem → detecting collisions is hard
- Security in MANETs
  - Routing denial of service: malicious or selfish nodes may disagree to route packets
  - Distributing false routing information by malicious nodes
  - Key distribution issue

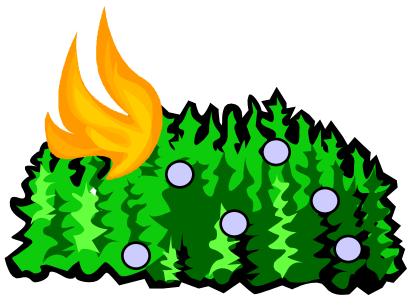
Wireless Sensor Networks (WSNs)

- WSNs consist of a large number of network sensors connected by wireless or wired networks

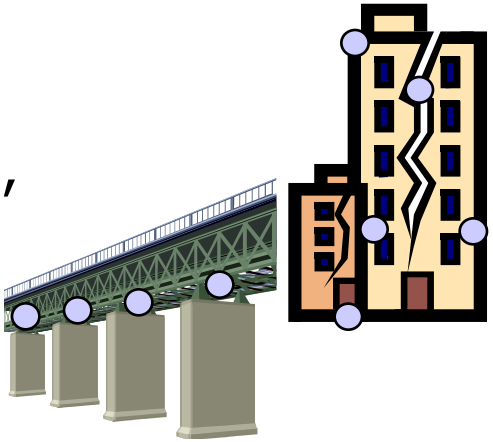
- ***Network sensors*** (or smart sensors, micro-sensors)
  - Sensor modules with low power and cheap sensors (e.g. acoustic, image, seismic, …) for environment sensing
  - Processors for signal processing, network protocol processing, data processing, etc.
  - RF communication modules

WSN application examples

- Disaster relief operations
  - Drop sensor nodes from an aircraft over a wildfire
  - Each node measures temperature
  - Derive a "temperature map"

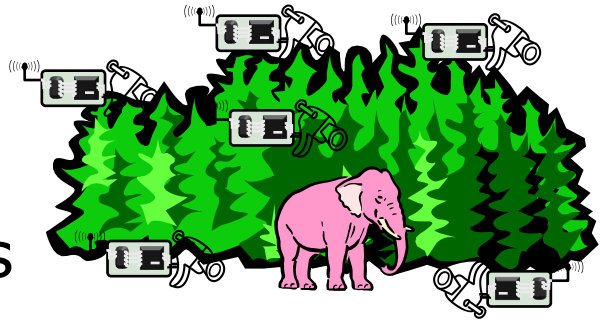


- Intelligent buildings (or bridges)
  - Reduce energy wastage by proper humidity, ventilation, air conditioning (HVAC) control
  - Needs measurements about room occupancy, temperature, air flow, …
  - Monitor mechanical stress after earthquakes



WSN application examples

- Habitat monitoring
  - Use sensor nodes to observe wildlife



- Machine monitoring and diagnosis
  - Embed sensing/control functions into places no cable has gone before
  - (e.g.) tire pressure monitoring
- Precision agriculture
  - Bring out fertilizer/pesticides/irrigation only when and where needed

WSN application examples

- ***Sources*** of data: measure data, report them "somewhere"
  - Typically equip with different kinds of actual sensors

- ***Sinks*** of data: interested in receiving data from WSN
  - Part of the WSN or external entity, PDA, gateway, …

WSN Application Types

- Based on the traffic pattern
  - ***Event detection:*** Nodes detect events, report these events to interested sinks
  - ***Periodic measurement:*** Nodes senses the environment, report the sensing data to interested sinks

- Deployment types
  - ***Random deployment: (e.g.)*** Dropped from aircraft (fixed)
  - ***Regular deployment: (e.g.)*** Well planned (fixed)
  - ***Mobile*** sensor nodes

MANETs vs. WSNs

MANETs	WSNs
Unreliable communication	Unreliable communication
Require self-configuration	Require self-configuration
Constrained energy and bandwidth	Very constrained energy and bandwidth
Small scale	Large scale
Typically mobile	Typically immobile
Competitive	Cooperative
Address centric	Data centric
QoS: delay, throughput, etc.	Application-specific QoS